In physics, **gravity** (from Latin *gravitas* 'weight'[1]) is a fundamental interaction which causes mutual attraction between all things with mass or energy. Gravity is, by far, the weakest of the four fundamental interactions, approximately 10^{38} times weaker than the strong interaction, 10^{36} times weaker than the electromagnetic force and 10^{29} times weaker than the weak interaction. As a result, it has no significant influence at the level of subatomic particles.[2] However, gravity is the most significant interaction between objects at the macroscopic scale, and it determines the motion of planets, stars, galaxies, and even light.

On Earth, gravity gives weight to physical objects, and the Moon's gravity is responsible for sublunar tides in the oceans (the corresponding antipodal tide is caused by the inertia of the Earth and Moon orbiting one another). Gravity also has many important biological functions, helping to guide the growth of plants through the process of gravitropism and influencing the circulation of fluids in multicellular organisms. Investigation into the effects of weightlessness has shown that gravity may play a role in immune system function and cell differentiation within the human body.

The gravitational attraction between the original gaseous matter in the Universe allowed it to coalesce and form stars which eventually condensed into galaxies, so gravity is responsible for many of the large-scale structures in the Universe. Gravity has an infinite range, although its effects become weaker as objects get farther away.